

List of Symbols

In this work, lowercase letters are generally used for scalar numbers and vectors; vectors are distinguished by scalars with an underline. Capital, Latin letters are used for matrices. Few exceptions to this rule are presents to maintain the commonly adopted capitals letters for some scalar quantities (the Young's modulus, for instance). Some special fonts are used to identify the work, the energy, and the potentials. An overline is also used when the quantity is evaluated in a special configuration, generally the reference one.

Operators

Δ	finite difference
d	differential and derivative symbol
∂	partial derivative symbol
$\dot{a}$	differentiation of a with respect to time
a'	differentiation of a with respect the curvilinear coordinate
$\ \underline{a}\ $	norm of the vector $\underline{a}$
a^*	predicted value of a in a predictor-corrector algorithm
t	transpose operator
$\wedge$	vector product
A^+	Moore-Penrose pseudoinverse of A

Indexes

α	index used to identify the component of a vector in the x_α -coordinate direction, with $\alpha = 1, 2, 3$
e	index of the e^{th} element, with $e = 1, \dots, m$
h	index of the h^{th} component of a column vector
i	index of the i^{th} node, with $i = 1, \dots, n$
j	index of the j^{th} node, with $j = 1, \dots, n$
k	index of the k^{th} time step
m	number of the elements
n	number of the nodes

Latin alphabet

0	null matrix, with $0 \in \mathbb{R}^{3 \times 3}$
a_e	cross section of the e^{th} element
$\bar{a}_e$	cross section of the e^{th} element in the reference configuration
A_e	assembly matrix of the e^{th} element, with $A_e \in \mathbb{R}^{6 \times 3n}$
D	damping matrix for the net, with $D \in \mathbb{R}^{3n \times 3n}$
d_t	distance between the centres of mass of the chaser satellite and the target, $d_t \in \mathbb{R}$
E	Young's modulus of the cable material, $E \in \mathbb{R}$
$\mathcal{E}_T$	total mechanical energy of the net
$\underline{e}_\alpha$	with $\alpha = 1, 2, 3$ - unit vectors in the direction of the x , y , and z axes, respectively, $\underline{e}_\alpha \in \mathbb{R}^3$
$\underline{e}_e$	unit vectors in the radial direction in a spherical coordinate system, $\underline{e}_r \in \mathbb{R}^3$
e_{11}	component of the Green-Lagrange strain tensor in the axial direction of the element
$\underline{f}$	vector of equality constraints, $\underline{f} \in \mathbb{R}^n$
$\underline{g}$	vector of inequality constraints, $\underline{g} \in \mathbb{R}^n$
I	identity matrix, with $I \in \mathbb{R}^{3 \times 3}$
J	The Jacobian matrix, with $J \in \mathbb{R}^{n \times 3n}$
$\mathcal{K}$	kinetic energy of the net
$\mathcal{L}$	Lagrangian function
l_e	nominal length of the e^{th} element, defined as the distance between its end nodes
$\bar{l}_e$	nominal length of the e^{th} element in the reference configuration
L_i	localisation matrix of the i^{th} node, with $L_i \in \mathbb{R}^{3 \times 3n}$
m_T	total mass of the net
$\underline{Q}$	linear momentum of the net, $\underline{Q} \in \mathbb{R}^3$
$\underline{q}$	generalized coordinates, $\underline{q} \in \mathbb{R}^{3n+2n}$
$\mathcal{R}$	Rayleigh dissipation function
$\mathbb{R}$	set of real numbers
$\underline{r}$	position vector with respect to the centre of the Earth, with $\underline{r} \in \mathbb{R}^{3n}$
$\Delta \underline{r}$	residuals, with $\Delta \underline{r} \in \mathbb{R}^{3n}$
r_t	radius of the spherical target, $r_t \in \mathbb{R}$
$\underline{s}$	slack variables, $\underline{s} \in \mathbb{R}^n$
S	secant elastic stiffness matrix for the net, $T \in \mathbb{R}^{3n \times 3n}$
S_e	secant elastic stiffness matrix for the e^{th} element, $S_e \in \mathbb{R}^{6 \times 6}$
t	time
T	tangent elastic stiffness matrix for the net, $T \in \mathbb{R}^{3n \times 3n}$
T_e	tangent elastic stiffness matrix for the e^{th} element, $T_e \in \mathbb{R}^{6 \times 6}$
$\mathcal{U}$	elastic potential energy of the net
$\mathcal{U}_e$	elastic potential energy of the e^{th} element

$\mathcal{V}$	potential of conservative forces
$\underline{v}$	vector of nodal velocities, with $\underline{v} \in \mathbb{R}^{3n}$
$\underline{v}_0$	vector of nodal velocities at time $t = 0$, with $\underline{v}_0 \in \mathbb{R}^{3n}$
$\underline{v}_G$	velocity of the centre of mass, with $\underline{x}_G \in \mathbb{R}^3$
$\mathcal{W}_d$	work of the damping forces
x_1	Cartesian coordinate
x_2	Cartesian coordinate
x_3	Cartesian coordinate
$\underline{x}$	vector of nodal positions, with $\underline{x} \in \mathbb{R}^{3n}$
$\underline{x}_0$	vector of nodal positions at time $t = 0$, with $\underline{x}_0 \in \mathbb{R}^{3n}$
$\underline{x}_e$	nodal position vector of the e^{th} element, with $\underline{x}_e \in \mathbb{R}^6$
$\underline{x}_G$	position vector of the centre of mass, with $\underline{x}_G \in \mathbb{R}^3$
$\underline{x}_i$	position vector of the i^{th} node, with $\underline{x}_i \in \mathbb{R}^3$

Greek alphabet

α_d	constant of proportionality with the mass for Rayleigh damping
β_d	constant of proportionality with the stiffness for Rayleigh damping
β	weighting parameter in the β -Newmark method
$\bar{\varphi}$	strain energy density evaluated in the reference configuration
γ	weighting parameter in the β -Newmark method
$\underline{\lambda}$	Lagrange multipliers, $\underline{\lambda} \in \mathbb{R}^n$
ν	Poisson's ratio
ν_i	modulus of the relative velocity of the i^{th} node with respect to the centre of mass of the net
ρ_e	density of the material constituting the e^{th} element
$\bar{\rho}_e$	density of the material constituting the e^{th} element, measured in the reference configuration
σ_{11}	PK2 axial stress in the cable
ξ_i	modulus of the relative distance of the i^{th} node with respect to the centre of mass of the net
Ω	current configuration of a body
$\bar{\Omega}$	reference configuration of a body

Acronyms

ADR	Active Debris Removal
ANCF	Absolute Nodal Coordinate Formulation
ESA	European Space Agency
FE	Finite Element
FE	Finite Element Analysis
GEO	Geostationary Orbit
GNC	Guidance, Navigation and Control
IADC	Inter-Agency Space Debris Coordination Committee
ISS	International Space Station
LEO	Low Earth Orbit
NASA	National Aeronautics and Space Administration
PFEF	Position-based Finite Element Formulation
PK2	Second Piola-Kirchhoff
TRL	Technology Readiness Level
UN COPUOS	United Nations Committee on Peaceful Uses of Outer Space
US SSN	United States Space Surveillance Network